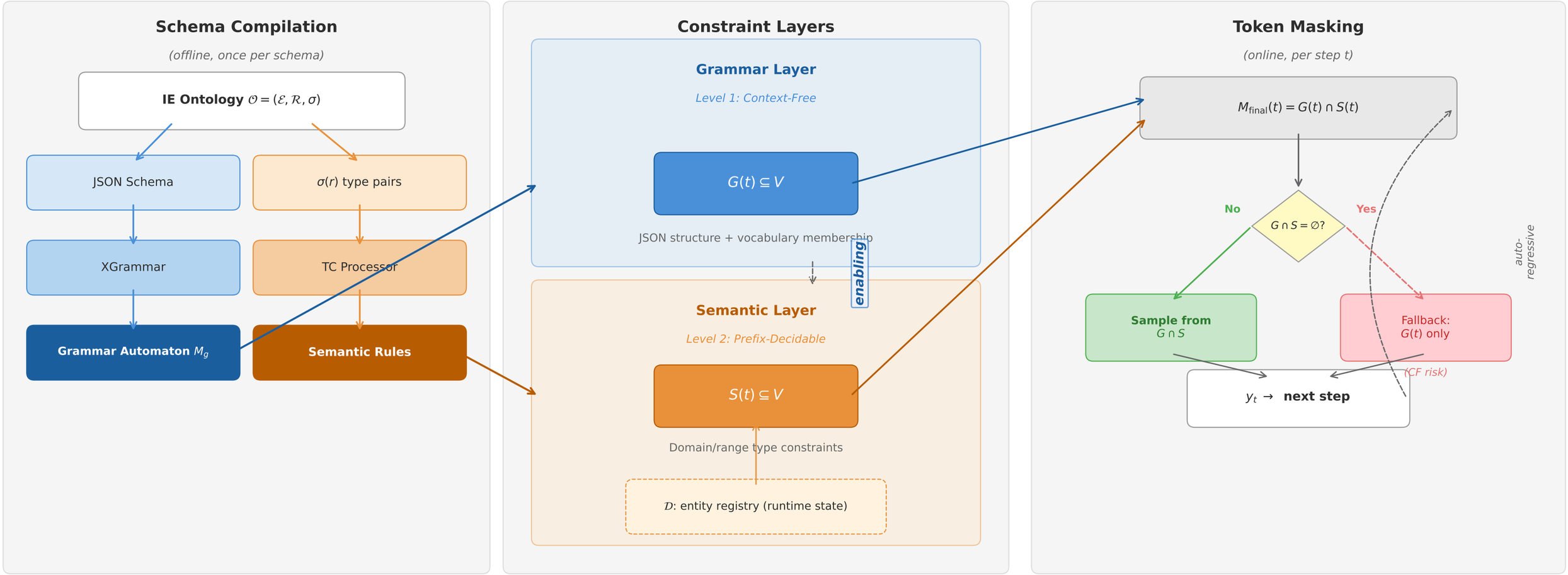
Schema Compilation
(offline, once per schema)
IE Ontology $\mathcal{O} = (\mathcal{E}, \mathcal{R}, \sigma)$
JSON Schema
$\sigma(r)$ type pairs
XGrammar
TC Processor
Grammar Automaton M_g
Semantic Rules
Constraint Layers
Grammar Layer
Level 1: Context-Free
$G(t) \subseteq V$
JSON structure + vocabulary membership
enabling
Semantic Layer
Level 2: Prefix-Decidable
$S(t) \subseteq V$
Domain/range type constraints
$\mathcal{D}$: entity registry (runtime state)
Token Masking
(online, per step t)
$M_{\text{final}}(t) = G(t) \cap S(t)$
No
$G \cap S = \emptyset$?
Yes
Sample from $G \cap S$
Fallback: $G(t)$ only
(CF risk)
$y_t \rightarrow$ next step
auto-regressive